AR
2EB

ALAMEDA RESEARCH
2: ELECTRIC BOOGALOO
LONG HORIZONS · STRUCTURAL SHIFTS · IMAGINATION

INTERNAL RESEARCH · MEMO · NOT INVESTMENT ADVICE · AI-ASSISTED

Illumina

NASDAQ : ILMN · Mature-Company DCF

$27.77B mkt cap · 151M sh · $1.16B cash, $1.99B net debt · ~$0.83B net debt; ~151M shares; ~$700M/yr buyback (no dividend); Revenue $4.34B FY25 (~flat 4yrs; Q1'26 +4.8% re-accel); ~23% non-GAAP op margin, ~70% gross, ~$0.9-1.0B FCF; Rallied ~80% off lows on the clinical/NovaSeq-X turnaround; ~5x EV/sales, ~22x FCF

2 July 2026 · the qualitative

$183.89

THE CENTRAL QUESTION

Is Illumina's ~80% rally + ~5x EV/sales pricing a durable clinical/NovaSeq-X re-acceleration — or is flat revenue + low-cost-entrant (Ultima/Element) + Roche erosion the reality?

Illumina trades at ~$22B (≈$145/sh, up ~80% off the lows) — ~5x EV/sales, ~22x FCF on $4.34B revenue (flat for four years, Q1'26 +4.8%), ~70% gross margin, ~$1B FCF. The mature DCF asks whether the NovaSeq-X consumables pull-through + clinical mix (>65% of sequencing consumables, +20% ex-China) re-accelerate the franchise toward a ~26% margin, against low-cost entrants (Ultima/Element $100 genome), Roche's 2026 SBX re-entry, China decline, and NIH cuts. The finding: the rally has priced the re-acceleration — only the bull-or-better clears spot; the base (modest re-accel) sits ~27% below. The TXG mirror.

PROBABILITY-WEIGHTED EXPECTED VALUE

$101.53 expected fair value today

-44.8% vs spot $183.89

FORWARD COMPOUNDED VALUE

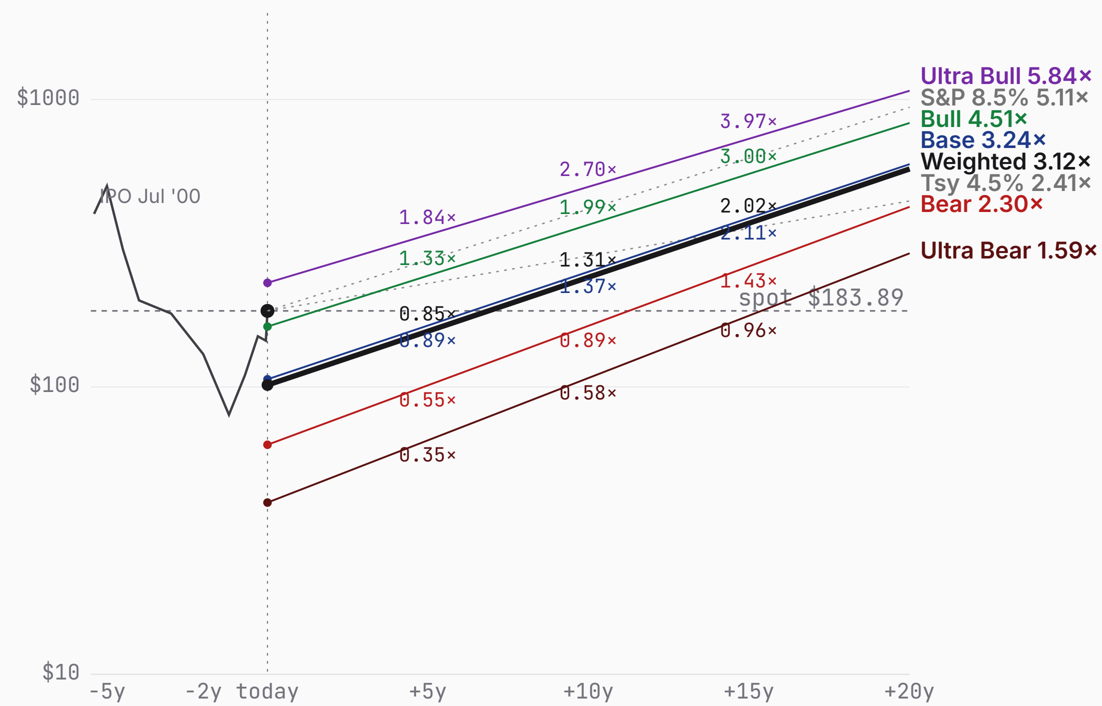
+5y	+10y	+15y	+20y
$156.27	$240.77	$371.34	$573.33
0.85× spot	1.31× spot	2.02× spot	3.12× spot

WEIGHTING RATIONALE

Ultra Bear 12% Entrants + China + NIH crush; ~$40 (-73%).
Bear 30% Flat; clinical offsets; ~$92 (-37%).
Base 35% Modest re-accel; ~26% margin; ~$106 (-27%).
Bull 18% Clinical compounds; multiomics; ~$162 (+12%).
Ultra Bull 5% Sequencing super-cycle; ~$230 (+59%).

Bull (18%) + Ultra Bull (5%) together contribute $40.70 — 40% of the $101.53 expected value despite 23% combined probability. Today's spot ($183.89) sits 81% above the weighted expected; forward-weighted value crosses spot between +5y and +10y.

PRICE + PROJECTIONS · HISTORICAL THROUGH PROBABILITY-WEIGHTED FORWARD



ULTRA BEAR	BEAR	BASE	BULL	ULTRA BULL
$39.59	$62.91	$106.30	$162.15	$230.21
-78.5% vs spot	-65.8% vs spot	-42.2% vs spot	-11.8% vs spot	+25.2% vs spot
CAGR -1.0% WACC 10.5%	CAGR +1.4% WACC 10.0%	CAGR +4.6% WACC 9.0%	CAGR +7.2% WACC 8.5%	CAGR +10.0% WACC 8.0%
CAGR -1.0% Prob 12%	CAGR +1.4% Prob 30%	CAGR +4.6% Prob 35%	CAGR +7.2% Prob 18%	CAGR +10.0% Prob 5%
Low-cost entrants + China + NIH crush.	Flat; clinical offsets erosion; rally unwinds.	Modest re-accel; NovaSeq-X pull-through; ~26% margin.	Clinical compounds; multiomics; margin >26%.	Sequencing super-cycle; dominant + multiomics.

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Illumina scenario narratives

PROBABILITY WEIGHTS

Ultra Bear 12% Bear 30% Base 35% Bull 18% Ultra Bull 5% · Weighted expected $101.53 (-44.8% vs spot)

ULTRA BEAR $39.59

-78.5% vs spot

Probability 12%

Low-cost entrants + China + NIH crush.

WHY 12%

The bear: the SBS patent estate ages, low-cost entrants design around it, and the academic base shrinks. 12% weight on real erosion.

WHAT HAPPENS

Ultima/Element ($100 genome) compress consumables pricing/share, Roche's SBX re-entry bites, China keeps declining, and NIH cuts freeze research demand. Revenue declines and the consumables annuity erodes.

DCF ~$40 (-73%) — a shrinking franchise losing its pricing Power earns a distressed multiple.

BEAR $62.91

-65.8% vs spot

Probability 30%

Flat; clinical offsets erosion; rally unwinds.

WHY 30%

Flat revenue (the FY22-25 reality) with clinical offsetting erosion. 30% weight — the rally priced more than this.

WHAT HAPPENS

Clinical growth roughly offsets China + entrant erosion, leaving revenue ~flat at a recovering margin; the franchise is dominant but ex-growth.

DCF ~$92 (-37%) — well below the rallied price; flat revenue doesn't justify ~5x sales.

BASE $106.30

-42.2% vs spot

Probability 35%

Modest re-accel; NovaSeq-X pull-through; ~26% margin.

WHY 35%

NovaSeq-X pull-through + clinical + the ~26% margin target, consistent with Q1'26's +4.8% and raised guide. 35% as the central outcome — but priced in.

WHAT HAPPENS

The modal path: the NovaSeq-X installed base matures into a consumables acceleration, clinical compounds, and the margin marches to ~26% (management's 2027 target). Revenue grows ~4-5%.

DCF ~$106 (-27%) — even a successful modest re-acceleration doesn't justify the ~80% rally; the entry has run ahead of the fundamentals.

BULL $162.15

-11.8% vs spot

Probability 18%

Clinical compounds; multiomics; margin >26%.

WHY 18%

The clinical-led re-acceleration is durable + multiomics adds growth. ~18%; the bull the rally is discounting in.

WHAT HAPPENS

Clinical sequencing compounds high-single-digits, multiomics/proteomics (SomaLogic) cross-sells, and the margin pushes past 26% as NovaSeq-X pull-through accelerates in 2H'26+.

DCF ~$162 (+12%) — the durable re-acceleration the rally is pricing actually materializes.

ULTRA BULL $230.21

+25.2% vs spot

Probability 5%

Sequencing super-cycle; dominant + multiomics.

WHY 5%

Sequencing super-cycle + ILMN defends its Power. ~5%, the asymmetric upside.

WHAT HAPPENS

A genomics super-cycle (clinical at scale + multiomics) re-accelerates ILMN to low-double-digits at ~30% margins, defending its ~80% share against entrants.

DCF ~$230 (+59%) — the dominant-platform-re-rates outcome; ~5% probability.

INTERNAL RESEARCH · MEMO · NOT INVESTMENT ADVICE · AI-ASSISTED

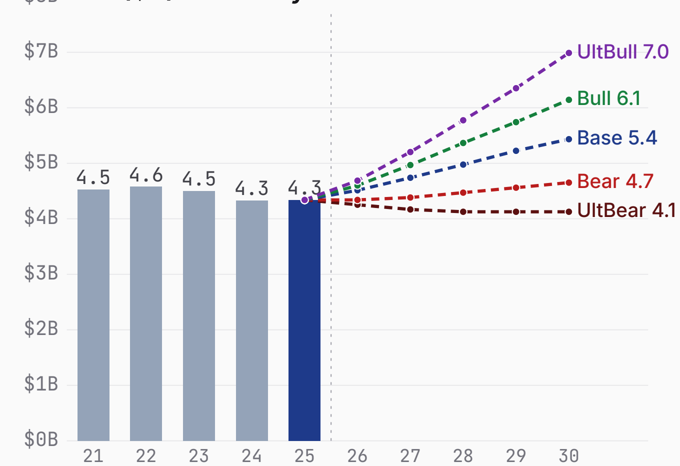
2 July 2026 · the snapshot

Illumina business snapshot

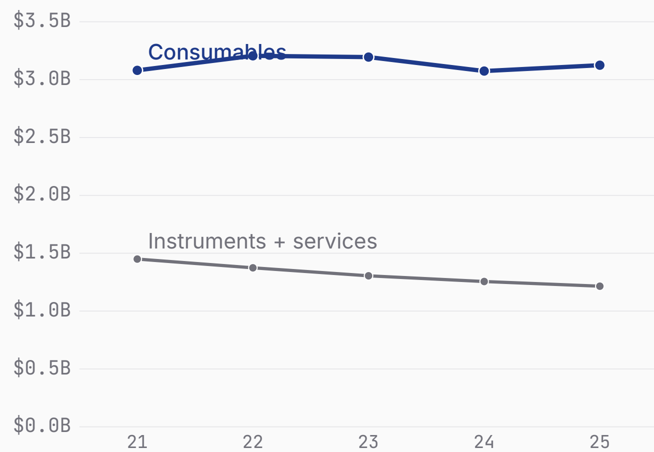
Bear $62.91 Base $106.30 Bull $162.15

FY21-FY25 history + FY26-FY30 scenario projections · calendar fiscal year · FY2025 10-K + Q1'26 (Core Illumina, post-GRAIL)

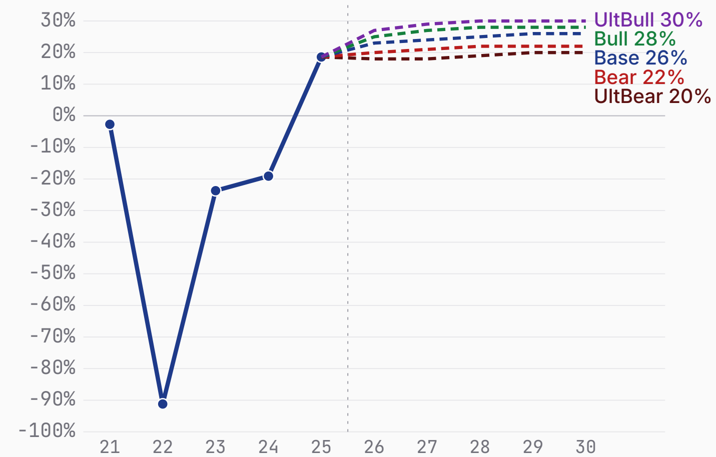
Revenue ($B) — history + scenarios to FY30



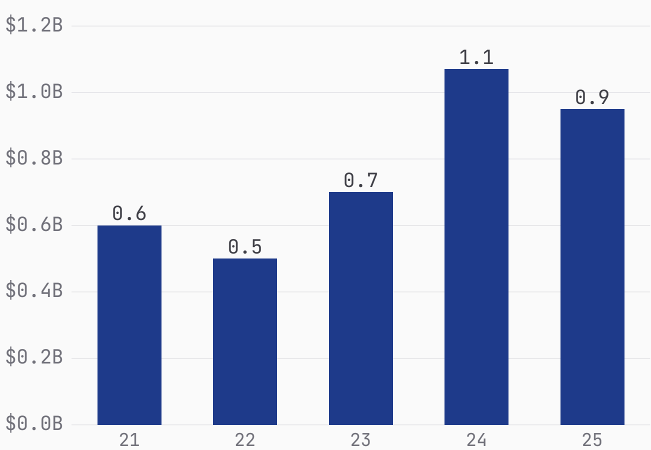
Revenue by segment ($B)



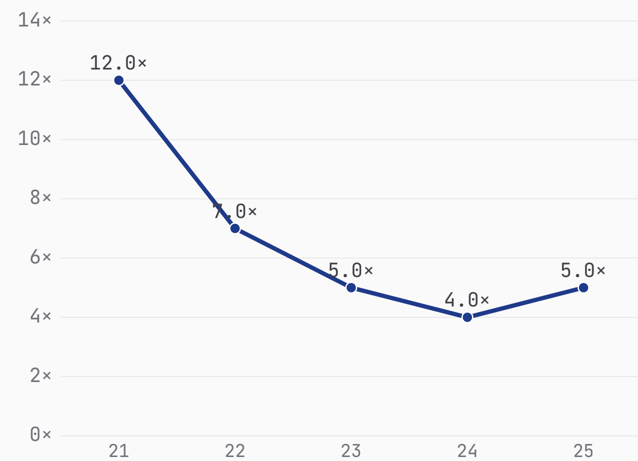
Op margin — history + scenarios



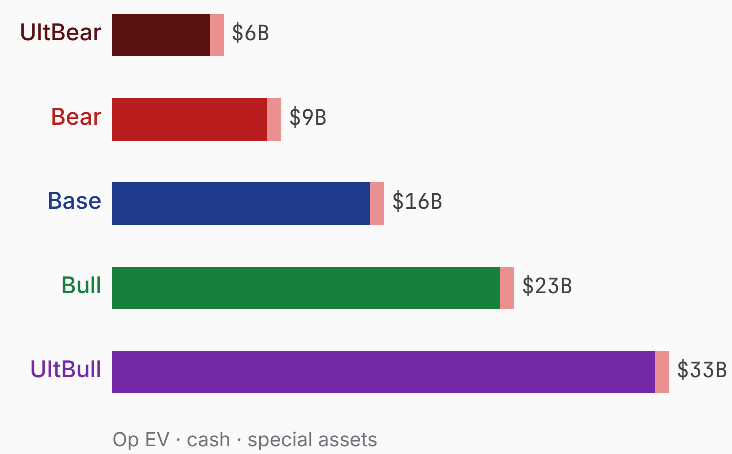
Free cash flow ($B) — history



EV / Revenue — historical



Equity build (Op EV - net debt)



Sources: Illumina FY2025 results (Core revenue $4.34B, ~23% non-GAAP op margin, ~$1B FCF, net debt ~$0.83B), Q1'26 (+4.8%, raised guide). GRAIL divested Jun 2024 → Core-only. Revenue as a growth-rate path off FY25; FCF = revenue x FCF margin; Gordon terminal. EV/sales vs life-science-tools peers.

2 July 2026 · the competitive landscape

Illumina 7 Powers · the moat, scored

Bear $62.91 Base $106.30 Bull $162.15

Arena. DNA sequencing — Illumina (~80% share, NovaSeq-X installed base, ~70%-margin consumables annuity) vs Ultima, Element, Roche (SBX, 2026), PacBio, Thermo, MGI/Complete Genomics; defending a scale + switching Power against low-cost entrants. (Supplies 10x Genomics/TXG; the two are IP litigants.)

POWER AUDIT
dominant-power durability: medium

POWER AUDIT · 7 POWERS

Scale Economies · thesis leans here (3/3)	~80% share + huge installed base → manufacturing/R&D cost advantage on consumables; the dominant Power.
Network Economies (1/3)	Published-protocol/citation network is mild.
Counter-Positioning (1/3)	Incumbent; low-cost entrants counter-position on price-per-genome.
Switching Costs (3/3)	Validated clinical assays + DRAGEN pipelines on Illumina chemistry won't re-validate cheaply (regulatory friction).
Branding (2/3)	The 'gold standard' in clinical NGS.
Cornered Resource (2/3)	Sequencing-by-synthesis patent estate — established but contested/aging (litigated vs Element/BGI).
Process Power (2/3)	Chemistry/manufacturing learning curve.

PEER SET

Ultima Genomics · private
UG 100, ~$80-100 genome (high-throughput) — the low-cost price attack.

Element Biosciences · private
AVITI + Vitari ($100 30x genome); mutual IP litigation with Illumina.

Roche (SBX) · diversified
Re-entering with sequencing-by-expansion (2026); diagnostics channel — the credible long-term threat.

PacBio / Thermo / MGI · varies
Long-read niche / Ion Torrent / China (MGI blocked from US by BIOSECURE).

THREAT VECTORS · FALSIFIERS

Scale Economies · Ultima / Element ($100 genome)
falsifier: Low-cost entrants take material high-throughput share; consumables pricing compresses.

Cornered Resource · Roche SBX (2026)
falsifier: Roche's SBX gains clinical traction, breaking the SBS franchise.

Switching Costs · NIH / China
falsifier: NIH cuts + China decline shrink the academic installed base the annuity depends on.

> **COMPETITIVE READ**
> Illumina's scale + switching-cost Power is genuine and durable — ~80% share, >90% of clinical genomics, a ~70%-margin consumables annuity, and clinical labs that won't cheaply re-validate. But the Power is being audited: revenue was flat for four years, China collapsed, NIH froze research demand, low-cost entrants (Ultima/Element $100 genome) attack pricing, and Roche's SBX re-enters in 2026. The ~80% rally to ~5x EV/sales has priced the NovaSeq-X/clinical re-acceleration; only the bull-or-better clears spot, and the modal base sits ~27% below — the entry, not the franchise, is the problem (the TXG mirror).

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Illumina show your work

Bear $62.91 Base $106.30 Bull $162.15 · Weighted $101.53 (-44.8%)

ULTRA BEAR $39.59

ASSUMPTIONS

5y revenue CAGR (%)	-1.0
Year-5 op margin (%)	20.0
WACC (%)	10.5
Terminal growth (%)	1.0
5y revenue CAGR (%)	-1.0
Starting revenue ($B)	4.34
Scenario probability (%)	12

5-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY26	4.25	+18%	+0.68	+0.61
FY27	4.17	+18%	+0.63	+0.51
FY28	4.13	+19%	+0.62	+0.46
FY29	4.13	+20%	+0.66	+0.44
FY30	4.13	+20%	+0.66	+0.40

Σ PV explicit FCF	+2.43
+ PV terminal (g 1.0%)	4.26

EQUITY BUILD · $B & PER SHARE

Operating EV	$6.69B
+ Cash & investments	$1.16B
– Net debt	$1.99B
= Equity value	$5.86B
FY30 shares (M)	148
= Expected per share	**$39.59**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$65.22	$107.45	$177.02	$291.63
0.35×	0.58×	0.96×	1.59×

BEAR $62.91

ASSUMPTIONS

5y revenue CAGR (%)	+1.4
Year-5 op margin (%)	22.0
WACC (%)	10.0
Terminal growth (%)	2.0
5y revenue CAGR (%)	+1.4
Starting revenue ($B)	4.34
Scenario probability (%)	30

5-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY26	4.34	+20%	+0.78	+0.71
FY27	4.38	+21%	+0.79	+0.65
FY28	4.47	+22%	+0.85	+0.64
FY29	4.56	+22%	+0.87	+0.59
FY30	4.65	+22%	+0.88	+0.55

Σ PV explicit FCF	+3.14
+ PV terminal (g 2.0%)	7.00

EQUITY BUILD · $B & PER SHARE

Operating EV	$10.14B
+ Cash & investments	$1.16B
– Net debt	$1.99B
= Equity value	$9.31B
FY30 shares (M)	148
= Expected per share	**$62.91**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$101.32	$163.17	$262.79	$423.23
0.55×	0.89×	1.43×	2.30×

BASE $106.30

ASSUMPTIONS

5y revenue CAGR (%)	+4.6
Year-5 op margin (%)	26.0
WACC (%)	9.0
Terminal growth (%)	2.5
5y revenue CAGR (%)	+4.6
Starting revenue ($B)	4.34
Scenario probability (%)	35

5-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY26	4.51	+23%	+0.90	+0.83
FY27	4.74	+24%	+0.99	+0.84
FY28	4.98	+25%	+1.09	+0.85
FY29	5.23	+26%	+1.15	+0.81
FY30	5.43	+26%	+1.20	+0.78

Σ PV explicit FCF	+4.10
+ PV terminal (g 2.5%)	12.25

EQUITY BUILD · $B & PER SHARE

Operating EV	$16.35B
+ Cash & investments	$1.16B
– Net debt	$1.99B
= Equity value	$15.52B
FY30 shares (M)	146
= Expected per share	**$106.30**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$163.56	$251.65	$387.20	$595.75
0.89×	1.37×	2.11×	3.24×

BULL $162.15

ASSUMPTIONS

5y revenue CAGR (%)	+7.2
Year-5 op margin (%)	28.0
WACC (%)	8.5
Terminal growth (%)	3.0
5y revenue CAGR (%)	+7.2
Starting revenue ($B)	4.34
Scenario probability (%)	18

5-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY26	4.60	+25%	+1.01	+0.93
FY27	4.97	+27%	+1.19	+1.01
FY28	5.37	+28%	+1.34	+1.05
FY29	5.74	+28%	+1.44	+1.03
FY30	6.14	+28%	+1.54	+1.02

Σ PV explicit FCF	+5.05
+ PV terminal (g 3.0%)	19.13

EQUITY BUILD · $B & PER SHARE

Operating EV	$24.18B
+ Cash & investments	$1.16B
– Net debt	$1.99B
= Equity value	$23.35B
FY30 shares (M)	144
= Expected per share	**$162.15**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$243.82	$366.62	$551.27	$828.92
1.33×	1.99×	3.00×	4.51×

ULTRA BULL $230.21

ASSUMPTIONS

5y revenue CAGR (%)	+10.0
Year-5 op margin (%)	30.0
WACC (%)	8.0
Terminal growth (%)	3.0
5y revenue CAGR (%)	+10.0
Starting revenue ($B)	4.34
Scenario probability (%)	5

5-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY26	4.69	+27%	+1.13	+1.04
FY27	5.20	+29%	+1.35	+1.16
FY28	5.78	+30%	+1.56	+1.24
FY29	6.35	+30%	+1.78	+1.31
FY30	6.99	+30%	+1.96	+1.33

Σ PV explicit FCF	+6.08
+ PV terminal (g 3.0%)	27.44

EQUITY BUILD · $B & PER SHARE

Operating EV	$33.52B
+ Cash & investments	$1.16B
– Net debt	$1.99B
= Equity value	$32.69B
FY30 shares (M)	142
= Expected per share	**$230.21**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$338.25	$497.01	$730.27	$1,073.00
1.84×	2.70×	3.97×	5.84×

Illumina People · Opportunity · Context · Deal

Bear $62.91 Base $106.30 Bull $162.15

Underwrite the position the way a venture investor underwrites a round — across all four legs, public or private. **People** is the leg scored here (observable only); Opportunity, Context and Deal cross-reference the rest of the memo.

PEOPLE · WHO RUNS / OWNS / FUNDS IT

Jacob Thaysen (CEO since Sept 2023)

2%
insider ownership

LOW
key-person risk

Capital allocation (realized)
Dominated by the GRAIL episode — a major observable value-destruction: bought GRAIL (~$8B, 2021) and CLOSED it without regulatory clearance, drawing an EU prohibition + an unprecedented €432M "gun-jumping" fine (later voided on jurisdiction, 2024) and a forced spin-off (June 2024 → GRAL) that took a ~$4B impairment + a ~$1B recapitalization. No dividend; small buybacks ($1.5B authorized Apr 2026); R&D-heavy.

Incentive alignment
Comp redesigned after a pay revolt: CEO target equity $10M is 70% PSUs / 30% RSUs, PSUs vesting on 3-year relative TSR (the make-whole grant made explicitly 70% TSR-contingent in response to stockholder feedback). Base $1M, target bonus 125%.

GOVERNANCE FLAGS

- single-class, declassified board (annual elections)
- 2023 Icahn proxy fight: chairman John Thompson ousted; Icahn won one board seat
- say-on-pay FAILED 2023 → recovered ~86.5% (2024) after the comp redesign
- board/chair overhaul since (Gottlieb non-exec chair 2025; Corvex Meister joined; Icahn fully exited)

THE FOUR LEGS

People (2 of 5)
observable composite · 1–5

Opportunity
The scenario distribution · the Page-3 business snapshot · the Arthur Indicator (ILMN 7.08)

Context
The 7 Powers analysis (Power Audit)

Deal
Open-market purchase = the deal: entry price vs. the scenario distribution (the −30% finding) + position sizing.

> **PEOPLE READ**
> The cohort's governance-turnaround story — and the weakest recent record: a board that closed an $8B deal without clearance (forced unwind, ~$4B impairment, an EU fine), failed say-on-pay (2023), and lost a proxy seat to Icahn. Being repaired — new CEO, independent chair, redesigned TSR-linked pay, say-on-pay recovered, activist oversight — but the track record is the observable flag.

Illumina supporting analysis · disclaimers · glossary

Bear $62.91 Base $106.30 Bull $162.15

PUSHBACK · WHY THE BASE CASE IS TOO HARSH

1 **~80% share, >90% of clinical genomics.**
Dominant installed base + consumables lock-in (validated clinical assays don't re-validate cheaply) — a real scale + switching Power.

2 **~70% gross margin, ~$1B FCF.**
A razor/blade annuity (consumables ~72% of revenue) with a clear path to a ~26% operating margin (2027 target).

3 **NovaSeq-X pull-through is coming.**
~890 installed (end-FY25); ~6-month pull-through lag → a 2H'26 consumables acceleration the base is just starting to price.

4 **Clinical mix is the durable engine.**
>65% of sequencing consumables, +20% ex-China — less funding-exposed than the academic/research side.

5 **GRAIL overhang is gone.**
Divested Jun 2024; Core-only reporting, EU fine/impairments behind it; new CEO running the cost-out + clinical pivot.

FALSIFICATION TRIGGERS

Bull validation
total revenue growth sustains >6% · NovaSeq-X consumables accelerate (2H'26) · non-GAAP op margin reaches ~26% · clinical stays +20% ex-China

Bear validation
revenue stays flat-to-down · Ultima/Element take material share · Roche SBX gains clinical traction · China/NIH drag deepens

Reframe needed
if a low-cost entrant or Roche SBX structurally breaks the SBS consumables-pricing Power, re-rate toward a commoditized-sequencing multiple

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Forward-looking statements.
Scenarios, valuations, projections, and any forward-looking statements involve substantial assumptions and uncertainty. Actual results may differ materially from those projected. Past performance is not indicative of future results. The model is a model; reality is not.

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GLOSSARY · CONCEPTS REFERENCED IN THE NARRATIVE

Consumables (razor/blade) — Reagents/flow-cells that run on the installed base (~72% of revenue, ~70% margin) — the recurring annuity and the moat.

NovaSeq X — Illumina's flagship high-throughput sequencer; its installed-base ramp drives the consumables pull-through (the re-acceleration mechanism).

Clinical mix — >65% of sequencing consumables, +20% ex-China — the durable, less-funding-exposed demand engine vs academic/research.

Low-cost entrants — Ultima (~$80-100 genome) + Element (Vitari) + Roche SBX (2026) attacking the sequencing-by-synthesis franchise on price.

FCF margin — Free cash flow / revenue (~21% FY25); the mature-DCF lever as the margin marches to the ~26% 2027 target.